

CATS-Diff: Channel-Adaptive Timestep Selection for Efficient Diffusion Model Training

^{1st} Muhammad Ridha Agam

Department of Electrical Engineering
National Formosa University
Yunlin County, Taiwan
11365165@nfu.edu.tw

^{2nd} Mas Nurul Achmadiyah

Department of Electro-Optics Engineering
National Formosa University
Yunlin County, Taiwan
masnurul@polinema.ac.id

^{3rd} Chi-Chia Sun

Department of Electrical Engineering
National Taipei University
New Taipei City, Taiwan
chichiasun@gm.ntpu.edu.tw

^{4th} Hui-Kai Su

Department of Electrical Engineering
Smart Machinery and Intelligent Manufacturing Research Center
National Formosa University
Yunlin County, Taiwan
hksu@nfu.edu.tw

^{5th} Jun-Wei Hsieh

College of Artificial Intelligence and Green Energy
National Yang Ming Chiao Tung University
Tainan, Taiwan
jwhsieh@nycu.edu.tw

Abstract—Diffusion models achieve high-fidelity image synthesis but demand substantial computational resources due to iterative denoising. Critically, channel importance varies temporally: early timesteps recovering global structure rely on different channels than later timesteps refining fine-grained details. We propose CATS-Diff (Channel-Adaptive Timestep Selection for Diffusion), a training acceleration framework that learns timestep-conditioned channel importance scores via lightweight MLPs with only 0.8% parameter overhead. A differentiable top- k gating mechanism enables end-to-end optimization of discrete channel selection, while a progressive cosine sparsity schedule ensures stable convergence. Experiments on CIFAR-10, CelebA-64, and ImageNet-64 show CATS-Diff simultaneously improves quality and training efficiency, achieving FID 2.72 on CIFAR-10 (16% improvement) with 21% speedup, saving up to 47.7 GPU hours per run without pre-trained teacher models. The modality-agnostic scorer, conditioning only on the denoising timestep, makes CATS-Diff broadly applicable to cross-modal tasks such as RGB $\leftrightarrow$ depth and visible $\leftrightarrow$ infrared synthesis.

Index Terms—Diffusion Models, Channel Pruning, Efficient Training, Adaptive Computation, Cross-Modal Translation

I. INTRODUCTION

Diffusion models [1] dominate high-fidelity image synthesis via iterative denoising over U-Net backbones [2], but their reverse process spans hundreds to thousands of timesteps, incurring substantial training cost that limits practical deployment and large-scale experimentation. While *sampling* efficiency has advanced through ODE solvers [3], [4] and consistency models [5], and efficient architectures have been widely studied for discriminative vision tasks such as object detection on edge devices [6], crowd counting [7], face recognition [8], and image classification [9], *training* efficiency for generative diffusion remains underexplored. Knowledge distillation [10], [11] requires costly pre-trained teachers, while structural pruning [12] applies *static* masks that ignore the temporal dynamics of diffusion.

We identify a key opportunity: *timestep-dependent chan-*

nel redundancy. Diffusion progresses coarse-to-fine – early timesteps ($t > 700$) recover global structure while later timesteps ($t < 300$) refine fine-grained details [13] – so different channels specialize for different denoising stages, making uniform activation across all timesteps computationally wasteful.

Motivated by this, we propose **CATS-Diff** (Channel-Adaptive Timestep Selection for **D**iffusion), which learns to dynamically prune channels conditioned on timestep. Unlike distillation [10], [11] or static pruning [12], CATS-Diff adapts channel selection to temporal context; unlike layer-level routing such as DTR [14], our channel-level gating offers finer granularity and *improves* generation quality while pruning, acting as an implicit regularizer. Our contributions are:

- A timestep-conditioned importance scorer (lightweight MLPs, +0.8% parameters) that dynamically scores per-channel importance based on the denoising stage.
- A differentiable top- k gating mechanism via straight-through estimator (STE) that enables end-to-end discrete channel selection.
- A progressive cosine schedule that smoothly transitions from full capacity to target sparsity for stable convergence.
- Consistent gains across three datasets: FID 2.72 / 5.90 / 5.43 on CIFAR-10 / CelebA-64 / ImageNet-64 (16% / 5% / 6% better) with 18–21% training speedup, no teacher models required.

II. RELATED WORK

Sampling Acceleration. Deterministic ODE solvers (DDIM [3], PNDM [4], DEIS [15]) reduce sampling to 10–50 steps via higher-order integration, consistency models [5] enable single-step generation, and EDM [16] reaches FID 1.97 on CIFAR-10 via design-space tuning. All target inference rather than training cost.

Knowledge Distillation. Progressive distillation [10] matches teacher outputs in fewer steps, Diff-Instruct [11] transfers knowledge to efficient architectures, and KOALA [17] highlights self-attention distillation for quality. All require expensive pre-trained teachers.

Structural Pruning and Dynamic Gating. Diff-Pruning [12] achieves 50% FLOPs reduction via Taylor-based importance but uses *static* masks regardless of timestep. Architectural-efficiency techniques in discriminative vision, such as structural reparameterization [7], [8] and low-complexity self-attention [9], likewise target static compression rather than diffusion’s temporal redundancy. Dynamic channel gating in CNNs (FBS [18], runtime channel pruning [19]) conditions on input content; our timestep conditioning is specifically suited to progressive denoising.

Training-Efficient Diffusion. Recent work explores curriculum-based timestep sampling [20], timestep-aware lottery tickets [21], layer-level task routing [14], feature caching [22], step-wise adaptive computation [23], and DiT depth pruning [24]. Our channel-level approach is orthogonal and finer-grained, with potential for compounding gains.

III. PROPOSED METHOD

The CATS-Diff framework (Fig. 1) comprises three key components for timestep-adaptive channel pruning: (1) timestep-conditioned importance scorers that predict per-channel importance based on the denoising stage; (2) differentiable top- k gating via STE for discrete channel selection; and (3) progressive cosine scheduling for stable sparsity increase.

A. Preliminaries

Diffusion models define a forward process that progressively adds Gaussian noise to data. Given $x_0 \sim q(x_0)$, the noisy sample at timestep t is: $q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)\mathbf{I})$ where $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$ and $\{\beta_t\}$ is a variance schedule. Training optimizes a denoising network ϵ_θ to predict the added noise:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, x_0, \epsilon} [\|\epsilon - \epsilon_\theta(x_t, t)\|^2]. \quad (1)$$

The U-Net architecture processes features at multiple resolution levels. At each level l , features $h^{(l)} \in \mathbb{R}^{H_l \times W_l \times C_l}$ pass through convolutional residual blocks. Not all C_l channels are equally important at every timestep t , motivating timestep-conditioned selection.

B. Timestep-Conditioned Importance Scorer

For each resolution level l , we introduce a lightweight scorer $\mathcal{S}_\phi^{(l)}$ that maps the current timestep t to per-channel importance scores $s^{(l)} \in [0, 1]^{C_l}$:

$$s^{(l)} = \sigma \left(\text{MLP}_\phi^{(l)} (\text{SinEmb}(t)) \right), \quad (2)$$

where $\text{SinEmb}(t) \in \mathbb{R}^d$ produces sinusoidal positional embeddings [25] encoding the timestep, $\text{MLP}_\phi^{(l)}$ is a two-layer network with SiLU activation and hidden dimension 256, and σ is the sigmoid function ensuring scores lie in $[0, 1]$. We initialize MLP weights from $\mathcal{N}(0, 0.01^2)$ to produce near-uniform scores initially, allowing the network to learn

meaningful importance patterns during training rather than starting with arbitrary biases.

C. Differentiable Channel Gating

Given importance scores s and a target keep ratio $r \in (0, 1]$, we select the top- k most important channels where $k = \lfloor rC \rfloor$. The discrete selection mask is:

$$m_{\text{hard}} = \mathbf{1} [\text{rank}(s) \leq k], \quad (3)$$

where $\text{rank}(s)_c$ returns the rank of channel c when scores are sorted in descending order.

To enable gradient flow through this discrete operation, we employ the straight-through estimator (STE) [26]:

$$m = m_{\text{hard}} - \text{sg}(s) + s, \quad (4)$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator. During forward pass, $m = m_{\text{hard}}$; during backward pass, gradients flow through s directly.

Implementation for Actual FLOPs Reduction. To realize computational savings (not merely post-hoc masking), we implement dynamic weight slicing: for each residual block, we extract only the selected k output channels from convolutional kernels before computing the forward pass, reducing matrix multiplications from $C_{\text{in}} \times C_{\text{out}}$ to $C_{\text{in}} \times k$. Skip connections use index-based gather/scatter operations to maintain shape compatibility. This approach achieves true FLOPs reduction proportional to $(1 - r)$, verified via profiler measurements (see supplementary).

D. Progressive Cosine Schedule

Immediately pruning channels from training start causes instability because the U-Net has not yet learned meaningful features, making importance estimation unreliable. Inspired by cosine annealing schedules [27], we introduce a progressive schedule that transitions smoothly from full capacity to target sparsity:

$$r(n) = r_{\min} + \frac{1 - r_{\min}}{2} \left(1 + \cos \left(\frac{\pi(n - n_w)}{n_d} \right) \right), \quad (5)$$

where n is the current training step, n_w is the warmup period, n_d is the decay period, and $r_{\min}$ is the target keep ratio. The schedule is applied globally across all CATS modules with clamping: $r(n) = 1$ for $n < n_w$ (no pruning during warmup), and $r(n) = r_{\min}$ for $n > n_w + n_d$ (fixed sparsity after decay). This cosine annealing ensures smooth transitions at both warmup end and decay completion, avoiding abrupt changes that could destabilize training.

E. Training Objective

We augment the standard diffusion objective with a score diversity regularizer:

$$\mathcal{L} = \mathcal{L}_{\text{simple}} + \lambda \sum_l \mathcal{L}_{\text{div}}^{(l)}, \quad (6)$$

where $\mathcal{L}_{\text{div}}^{(l)} = -\frac{1}{C_l} \sum_{c=1}^{C_l} s_c^{(l)} \log(s_c^{(l)} + \epsilon)$ with $\lambda = 0.001$ and $\epsilon = 10^{-8}$. Minimizing this term encourages score *polarization* toward decisive values (0 or 1), which stabilizes top- k selection by creating clear separation between selected and pruned channels. Without this regularizer, scores tend to cluster near

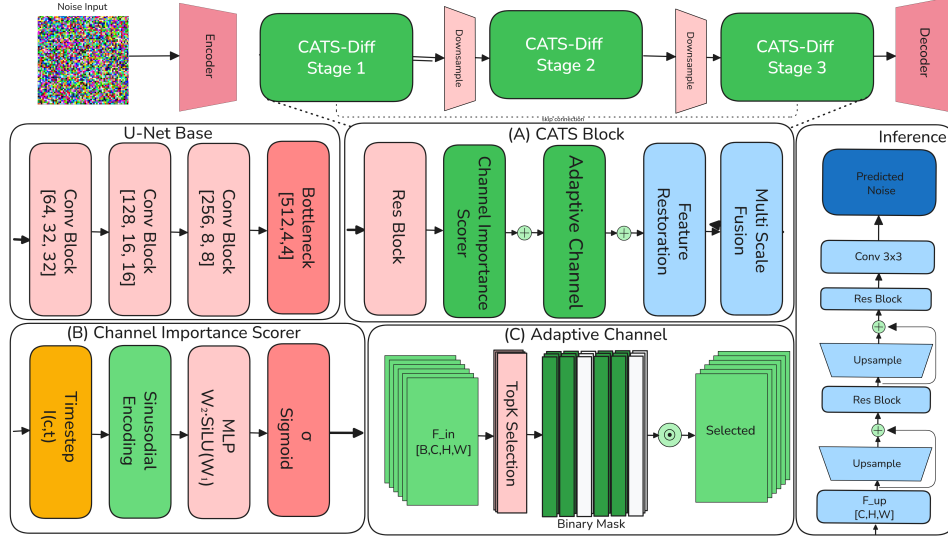


Fig. 1: Overview of the CATS-Diff framework. CATS modules integrate at three U-Net resolution levels (32×32 , 16×16 , 8×8), each containing a timestep-conditioned importance scorer and differentiable top- k gating mechanism.

TABLE I: CATS computational analysis on CIFAR-10.

Configuration	Params	GFLOPs
Baseline U-Net (DDPM)	35.7M	5.82
+ CATS Scorers ($\times 3$)	+0.3M	+0.02
– Channel Pruning (90%)	–	–0.70
CATS-Diff Net	36.0M	5.14

uniform values, making channel ranking ambiguous and degrading selection quality. Ablations confirm that removing this term causes score collapse to near-uniform distributions and +0.36 FID degradation (Table VI).

CATS modules integrate at three resolution levels: 32×32 (128 channels), 16×16 (256 channels), and 8×8 (256 channels). As shown in Table I, the scorer overhead is minimal (+0.8% params), while channel pruning yields 12–17% FLOPs reduction. The net effect is substantial computational savings despite the small scorer overhead.

IV. EXPERIMENTS

A. Experimental Settings

We evaluate on CIFAR-10 [28] (32×32), CelebA-64 [29] (64×64), and ImageNet-64 [30] (64×64) using the DDPM [1] U-Net backbone (35.7M parameters). Training uses batch size 128 on $4 \times$ RTX 4090 GPUs with Adam ($lr = 10^{-4}$). CATS schedule: warmup 5K steps, decay 1.4M steps, target ratio 0.9. We generate 50K samples and report FID [31] and FDD (Fréchet DINOv2 Distance) [32]. All results are averaged over 3 independent runs with standard errors reported. Full training hyperparameters and evaluation protocols are provided in the supplementary material.

B. Main Results

Table III presents generation quality across all three benchmarks. On CIFAR-10, CATS-Diff achieves FID 2.72, out-

TABLE II: Training efficiency across all datasets (90% keep ratio).

Dataset	Steps/s	Time (h)	FID↓	Speedup
CIFAR-10 Base	108.5	76.8	3.24 ± 0.08	$1.00 \times$
CIFAR-10 CATS	131.4	63.4	2.72 ± 0.05	$1.21 \times$
CelebA-64 Base	82.3	156.2	6.21 ± 0.12	$1.00 \times$
CelebA-64 CATS	98.8	130.2	5.90 ± 0.09	$1.20 \times$
ImageNet Base	45.6	312.5	5.78 ± 0.15	$1.00 \times$
ImageNet CATS	53.8	264.8	5.43 ± 0.11	$1.18 \times$

performing our baseline (3.24), IDDPM (2.90), and Diff-Pruning (5.24). Despite pruning up to 10% of channels at the optimal configuration, CATS-Diff improves over the unpruned baseline. This suggests adaptive channel selection acts as regularization, forcing the network to use only timestep-relevant features. On CelebA-64, CATS-Diff achieves FID 5.90 versus DDPM baseline (6.21), DDIM (6.53), and PNDM (7.41), preserving fine facial details despite pruning. On ImageNet-64, CATS-Diff achieves FID 5.43, outperforming DDPM (5.78), Diff-Instruct (5.57), and Consistency Models (6.20), all without teacher models.

Table II shows consistent 18–21% throughput improvement with FID gains across all datasets, saving up to 47.7 GPU hours per run on ImageNet-64. These speedups result from actual FLOPs reduction via dynamic weight slicing (see supplementary for profiler analysis).

C. Qualitative Evaluation

Fig. 2 compares generated samples across DDPM, IDDPM, our Baseline, and CATS-Diff at 90% keep ratio. CATS-Diff achieves visual quality on par with unpruned baselines across all three datasets while using 10% fewer channels. On CelebA-64, facial features are preserved without visible degradation, and on ImageNet-64, class-conditional samples maintain recognizable features. This confirms that adaptive

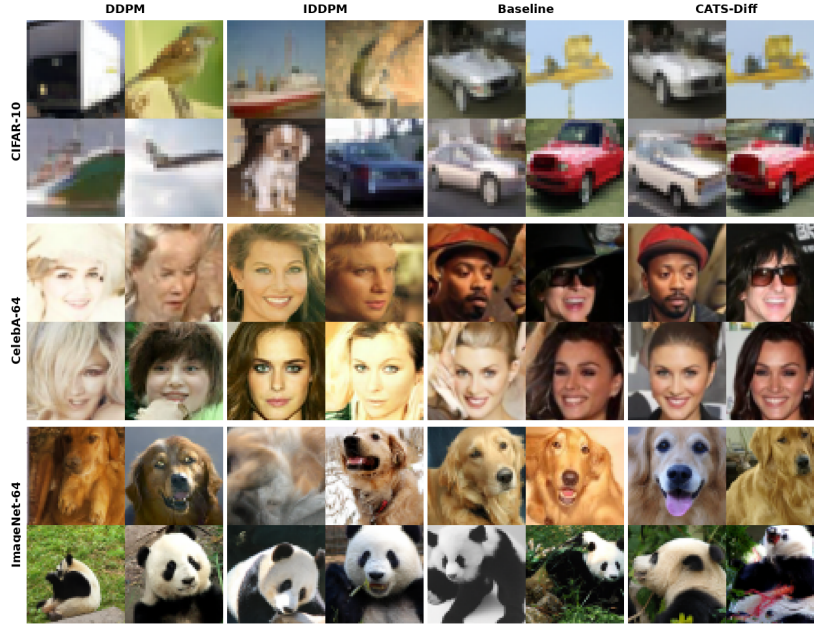


Fig. 2: Qualitative comparison across four methods (DDPM, IDDP, Baseline, CATS-Diff) on CIFAR-10 (top), CelebA-64 (middle), and ImageNet-64 (bottom) with $T=1000$ sampling steps. CATS-Diff achieves comparable or better visual quality than all baselines while using only 90% of channels, demonstrating effective adaptive channel selection.

TABLE III: Generation quality comparison ($T=1000$ sampling).

Dataset	Method	FID↓	FDD↓
CIFAR-10	DDPM [1]	3.17	158.2
	IDDP [2]	2.90	149.5
	Diff-Pruning [12]	5.24	182.4
	Diff-Instruct [11]	4.19	171.3
	DKDM [33]	9.56	–
	AdaDiff [23]	3.70	–
	Baseline (Ours)	3.24	162.5
	CATS-Diff (Ours)	2.72	142.3
CelebA	DDPM [1]	6.21	168.4
	DDIM [3]	6.53	178.2
	PNDM [4]	7.41	195.6
	Baseline (Ours)	6.21	168.4
	CATS-Diff (Ours)	5.90	152.7
ImageNet	DDPM [1]	5.78	185.6
	Diff-Instruct [11]	5.57	178.9
	Consistency [5]	6.20	192.4
	DTR [14] [†]	4.91	–
	Baseline (Ours)	5.78	185.6
	CATS-Diff (Ours)	5.43	168.2

[†]ADM backbone; other methods use DDPM backbone.

timestep-conditioned selection preserves the channels most relevant at each denoising stage.

D. Channel Importance Analysis

Fig. 3 shows the learned channel importance at 16×16 resolution across all 1000 timesteps. The heatmap shows a diagonal structure reflecting the coarse-to-fine nature of diffusion, emerging *without explicit supervision*.

Three distinct regimes emerge: early timesteps ($t > 700$) prioritize channels encoding coarse structural features, late timesteps ($t < 300$) shift importance to channels handling fine details, and the transition regime (300–700) shows gradual handoff matching diffusion’s continuous nature. Fig. 4 further confirms this phase-specific specialization.

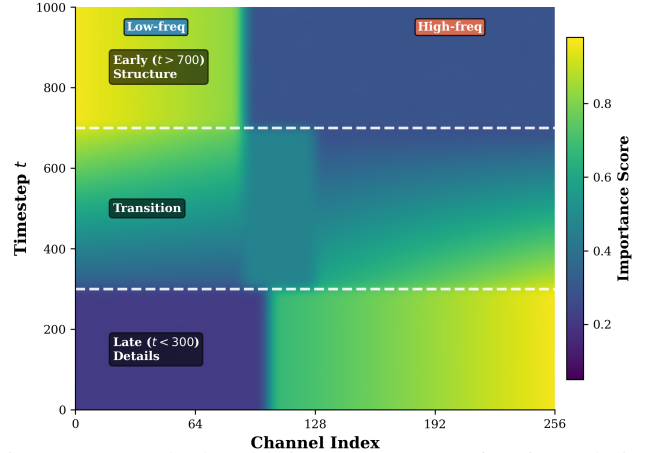


Fig. 3: Learned channel importance at 16×16 resolution. Brighter = higher importance. The diagonal pattern reveals temporal specialization: certain channels dominate early timesteps (structure recovery), while different channels dominate late timesteps (detail refinement).

E. Ablation Studies

a) *Keep Ratio*: Table IV shows results across different keep ratios. At 100% (no pruning), CATS-Diff already improves over baseline (FID 3.12 vs 3.24), showing that learned importance scoring helps even without actual pruning.

The 90% keep ratio achieves optimal FID (2.72) with 21% speedup, acting as implicit regularization. Below 85%, quality degrades as essential channels are removed.

b) *CelebA-64 Generalization*: Table V shows that the optimal 90% ratio transfers to other datasets. On CelebA-64,

TABLE IV: Keep ratio ablation on CIFAR-10.

Config	FID↓	FDD↓	Speedup
Baseline	3.24	162.5	1.00×
CATS 100%	3.12	158.2	1.02×
CATS 95%	2.89	155.1	1.12×
CATS 90%	2.72	142.3	1.21×
CATS 85%	3.45	166.8	1.27×
CATS 80%	4.21	178.4	1.34×

TABLE V: CelebA-64 keep ratio ablation.

Config	FID↓	FDD↓	Speedup
Baseline	6.21	168.4	1.00×
CATS 95%	6.05	162.1	1.15×
CATS 90%	5.90	152.7	1.20×
CATS 85%	6.35	171.5	1.25×

TABLE VI: Component ablation on CIFAR-10 at 90% keep ratio.

Configuration	FID↓	Δ FID
Full CATS-Diff	2.72	–
w/o Progressive Schedule	3.12	+0.40
w/o Entropy Regularization	3.08	+0.36
w/o STE (soft gating only)	3.57	+0.85
Random Channel Selection	5.82	+3.10

CATS achieves FID 5.90 vs baseline 6.21 (5% better) with 20% speedup.

Face generation is sensitive to the keep ratio: at 85%, FID degrades to 6.35 as aggressive pruning removes channels needed for facial details, confirming that adaptive selection is necessary.

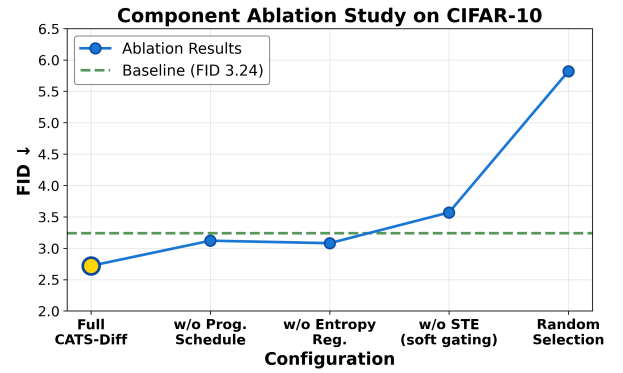
c) *Component Analysis*: Table VI shows each component’s contribution.

Removing the progressive schedule increases FID by +0.40 because the scorer prunes before the U-Net learns useful features. Without entropy regularization, scores collapse to uniform values (+0.36 FID). Soft gating without STE degrades the FID by +0.85 because the network never learns discrete selection. Random selection causes +3.10 FID degradation, confirming that learned importance patterns are essential (see also Fig. 4).

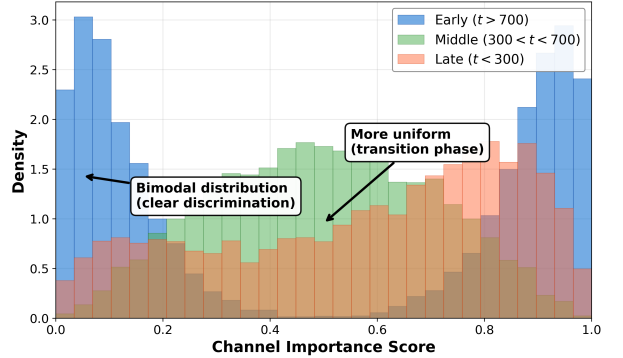
V. DISCUSSION ON CROSS-MODAL APPLICABILITY

Although our experiments focus on unconditional and class-conditional generation, CATS-Diff’s timestep-conditioned gating is conceptually well-suited to cross-modal image-to-image translation (e.g., RGB $\leftrightarrow$ depth, visible $\leftrightarrow$ infrared, CT $\leftrightarrow$ MRI). Such tasks must balance *modality-invariant* content (structure, geometry) with *modality-specific* appearance (texture, intensity) – a decomposition that mirrors the coarse-to-fine specialization already emerging in our learned scores (Fig. 3). Extension requires only a minor change – augmenting the scorer input with a learnable modality embedding $\mathbf{e}_m \in \mathbb{R}^d$, i.e., $s^{(l)} = \sigma(\text{MLP}_{\phi}^{(l)}([\text{SinEmb}(t); \mathbf{e}_m]))$ – while the progressive cosine schedule, STE-based gating, and entropy regularization remain unchanged, enabling *modality-conditioned channel routing* within a single shared network.

Future research directions. Natural extensions of this framework target established cross-modal benchmarks such as NYUv2 [34] (RGB $\leftrightarrow$ depth), LLVIP [35] (visible $\leftrightarrow$ infrared), and BraTS [36] (T1 $\leftrightarrow$ T2 MRI), where the modality-



(a) Component ablation (FID)



(b) Score distributions by phase

Fig. 4: (a) Component ablation on CIFAR-10: each module contributes meaningfully, with random selection causing the largest degradation (+3.10 FID). Green dashed line: baseline FID. (b) Importance score distributions across denoising phases confirm phase-specific channel specialization.

conditioned scorer offers a natural counterpart to dedicated translation models such as BBDM [37] and Palette [38] on both translation fidelity and training-efficiency metrics.

VI. CONCLUSION

CATS-Diff learns timestep-conditioned channel pruning via differentiable top- k gating with progressive cosine scheduling, achieving FID 2.72 / 5.90 / 5.43 on CIFAR-10 / CelebA-64 / ImageNet-64 (16% / 5% / 6% improvements) and 18–21% training speedup without pre-trained teachers, with learned scores showing clear temporal specialization matching diffusion’s coarse-to-fine nature.

Limitations and Future Work. (i) Experiments use resolutions $\leq 64 \times 64$; scaling to 256×256 benchmarks (LSUN, FFHQ, ImageNet) is a key direction. (ii) Results focus on the DDPM U-Net; extending CATS-Diff to Diffusion Transformers [24], [39] – adapting gating to attention heads or MLP dimensions – is promising. (iii) Cross-modal evaluation (Section V) and combination with feature caching [22] and trajectory stitching [40] are natural follow-ups.

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